

MATH – 241
Engineering Mathematics I
Problem Set – 7

Question 1. Show that

$$\det(A) = \frac{1}{2} \begin{vmatrix} \text{tr}(A) & 1 \\ \text{tr}(A^2) & \text{tr}(A) \end{vmatrix}$$

for every 2×2 matrix.

(**Note :** Trace of an $n \times n$ matrix $A = [a_{ij}]_{n \times n}$ is defined as the sum of the main diagonal entries i.e. $\text{tr}(A) = \sum_{i=1}^n a_{ii}$).

Question 1-Ans. For a 2×2 matrix $A = [a_{ij}]_{2 \times 2}$;

$$\det(A_{2 \times 2}) = a_{11}a_{22} - a_{21}a_{12} \text{ and } \text{tr}(A_{2 \times 2}) = \sum_{i=1}^2 a_{ii} = a_{11} + a_{22}.$$

If $B = A^2$ then;

$$\det(A) = \frac{1}{2} \begin{vmatrix} \text{tr}(A) & 1 \\ \text{tr}(B) & \text{tr}(A) \end{vmatrix} = \frac{1}{2} \begin{vmatrix} a_{11} + a_{22} & 1 \\ b_{11} + b_{22} & a_{11} + a_{22} \end{vmatrix}.$$

where $b_{11} = a_{11}^2 + a_{12}a_{21}$ and $b_{22} = a_{22}^2 + a_{12}a_{21}$. Then

$$\begin{aligned} \det(A) &= \frac{1}{2} \begin{vmatrix} \text{tr}(A) & 1 \\ \text{tr}(A^2) & \text{tr}(A) \end{vmatrix} = \frac{1}{2} ((a_{11} + a_{22})^2 - (a_{11}^2 + a_{12}a_{21} + a_{22}^2 + a_{12}a_{21})) \\ &= \frac{1}{2} ((a_{11} + a_{22})^2 - (a_{11}^2 + 2a_{12}a_{21} + a_{22}^2)) \\ &= \frac{1}{2} ((a_{11}^2 + 2a_{11}a_{22} + a_{22}^2) - (a_{11}^2 + 2a_{12}a_{21} + a_{22}^2)) \\ &= \frac{1}{2} (2a_{11}a_{22} - 2a_{12}a_{21}) = a_{11}a_{22} - a_{12}a_{21} = \det(A). \end{aligned}$$

Question 2. What is the maximum number of zeros that a 3×3 matrix can have without having a zero determinant? Explain your reasoning.

Question 2-Ans. - For a square matrix which has a column or row of zeros, then determinant of the matrix is zero. Therefore, for a 3×3 matrix we need at least 3 nonzero entries.

- For a triangular matrix, its determinant is the multiplication of its diagonal entries.
- A diagonal matrix is a triangular matrix. (Both upper and lower triangular.)
- For $A = [a_{ij}]_{3 \times 3}$ diagonal matrix determinant can be written as:

$$\det(A_{3 \times 3}) = \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33}.$$

- Therefore, number of possible zeros for a non-zero 3×3 matrix determinant is 6.
- For a $n \times n$ matrix with non-zero determinant, number of possible zero entries is $n^2 - n$.

Question 3. Prove that (x_1, y_1) , (x_2, y_2) , (x_3, y_3) are collinear points if and only if

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Question 3-Ans. ($\Rightarrow$) If the points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) are collinear, they should be lying on the same line. For a line equation $ax + by + c = 0$ all of the points should satisfy this equation. Then we can obtain the following linear system of equations that:

$$\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Augmented matrix of the linear system can be written as:

$$\left[\begin{array}{ccc|c} x_1 & y_1 & 1 & 0 \\ x_2 & y_2 & 1 & 0 \\ x_3 & y_3 & 1 & 0 \end{array} \right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc|c} x_1 & y_1 & 1 & 0 \\ x_2-x_1 & y_2-y_1 & 0 & 0 \\ x_3-x_1 & y_3-y_1 & 0 & 0 \end{array} \right] \xrightarrow{\frac{R_2}{x_2-x_1}} \left[\begin{array}{ccc|c} x_1 & y_1 & 1 & 0 \\ 1 & \frac{y_2-y_1}{x_2-x_1} & 0 & 0 \\ 1 & \frac{y_3-y_1}{x_3-x_1} & 0 & 0 \end{array} \right]$$

Then, $\frac{y_2-y_1}{x_2-x_1} = \frac{y_3-y_1}{x_3-x_1}$ which is the gradient of the line. Then second and third rows are the same resulting with zero determinant.

($\Leftarrow$) If the determinant is zero, the linear system:

$$\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

should have a non-trivial solution meaning that:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

If the system has a non-trivial solution, it means that there exists a line passing through all these points or these points are collinear.

Question 4. Prove that the equation of the line through the distinct points (a_1, b_1) and (a_2, b_2) can be written as:

$$\begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = 0.$$

Question 4-Ans. The equation of the line through the distinct points (a_1, b_1) and (a_2, b_2) can be written as;

$$\frac{x-a_1}{a_2-a_1} = \frac{y-b_1}{b_2-b_1}.$$

which can be also written as;

$$\frac{x-a_1}{a_2-a_1} = \frac{y-b_1}{b_2-b_1} \Rightarrow (a_1-x)(b_2-b_1) - (a_2-a_1)(b_1-y) = 0.$$

This line equation can be written as a 2×2 matrix determinant or a cofactor of a 3×3

matrix determinant defined as;

$$\begin{vmatrix} x & y & 1 \\ a_1 - x & b_1 - y & 0 \\ a_2 - a_1 & b_2 - b_1 & 0 \end{vmatrix} = 0.$$

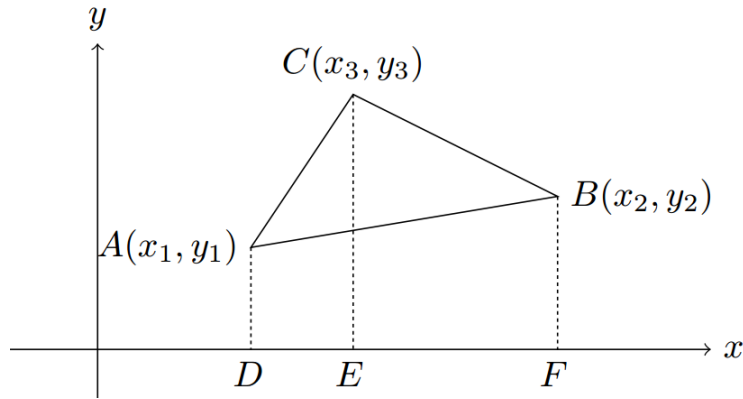
By applying elementary row operations we can obtain the following:

$$\begin{vmatrix} x & y & 1 \\ a_1 - x & b_1 - y & 0 \\ a_2 - a_1 & b_2 - b_1 & 0 \end{vmatrix} = 0 \xrightarrow{R_2+R_1} \begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 - a_1 & b_2 - b_1 & 0 \end{vmatrix} = 0 \xrightarrow{R_3+R_2} \begin{vmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{vmatrix} = 0.$$

which is the desired determinant of the line through the distinct points (a_1, b_1) and (a_2, b_2) .

Question 5. In the below figure, the area of the triangle ABC can be expressed as:

$$\text{area}(ABC) = \text{area}(ADEC) + \text{area}(CEFB) - \text{area}(ADFB).$$



Use this and the fact that area of a trapezoid equals $\frac{1}{2}$ the altitude times the sum of the parallel sides to show that:

$$\text{area}(ABC) = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

(**Note :** In the definition of this formula, the vertices are labeled such that the triangle is traced counter-clockwise proceeding from (x_1, y_1) to (x_2, y_2) to (x_3, y_3) . For a clockwise orientation, the determinant above yields the negative of the area.)

Question 5-Ans. Lets calculate the $area(ABC)$ by using trapezoid area formulation:

$$\begin{aligned}
area(ABC) &= area(ADEC) + area(CEFB) - area(ADFB) \\
&= \frac{1}{2}(y_1 + y_3)(x_3 - x_1) + \frac{1}{2}(y_3 + y_2)(x_2 - x_3) - \frac{1}{2}(y_1 + y_2)(x_2 - x_1) \\
&= \frac{1}{2}(y_1x_3 - y_1x_1 + y_3x_3 - y_3x_1 + y_3x_2 - y_3x_3 + y_2x_2 - y_2x_3 \\
&\quad - y_1x_2 + y_1x_1 - y_2x_2 + y_2x_1) \\
&= \frac{1}{2}(y_1x_3 - y_3x_1 + y_3x_2 - y_2x_3 - y_1x_2 + y_2x_1) \\
&= \frac{1}{2}((1)(x_2y_3 - x_3y_2) + (-1)(x_1y_3 - x_3y_1) + (1)(x_1y_2 - x_2y_1)) \\
&= \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.
\end{aligned}$$

Question 6. If A is invertible $n \times n$ matrix, prove that:

$$det(adj(A)) = det(A^{n-1}).$$

Question 6-Ans. If A is invertible $n \times n$ matrix, then A^{-1} exists and can be written as

$$A^{-1} = \frac{adj(A)}{det(A)} \Rightarrow adj(A) = det(A)A^{-1}.$$

if we multiply both sides with A , we obtain:

$$A.adj(A) = det(A).I.$$

if we take the determinant of both sides, we obtain:

$$det(A.adj(A)) = det(det(A).I) \Rightarrow det(A)det(adj(A)) = det(A)^n det(I)$$

which will result with the following:

$$det(adj(A)) = \frac{det(A)^n}{det(A)} = det(A)^{n-1} = det(A^{n-1}).$$